Midterm Bonus Assignment

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Abstract

This document serves as the solution to the Midterm Bonus Assignment problems.

Introduction

Euler showed that the sum of the inverses of the squares of the positive integers converges to $\pi^2/6$:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The purpose of this assignment is to prove this fact using methods from single and multivariable calculus.

Problem 1

Prove that the double integral and the infinite series shown below are equal:

$$\int_0^1 \int_0^1 \frac{1}{1-xy}\, dx\, dy = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

For the double integral, we have:

$$\int_0^1 \int_0^1 \frac{1}{1-xy}\, dx\, dy = \int_0^1 \left[\ln(1-xy)\right]_0^1 dy = \int_0^1 \ln(1-y)\, dy = \left[-y\ln(1-y)+y\right]_0^1 = 1.$$

Expanding the integrand into an infinite series, we have:

$$\frac{1}{1-xy} = \sum_{n=0}^{\infty}(xy)^n = \sum_{n=0}^{\infty} x^n y^n.$$

Problem 2

Conclusion